

The Influence of Metaverse Environment Design on Learning Experiences in Virtual Reality Classes: A Comparative Study

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Figure 1: We designed an experiment to compare four metaverses as meeting places for educational purposes. We selected the following commercially available metaverses: a. Arthur, b. Mozilla Hubs, c. Spatial, and d. Meta Workrooms.

ABSTRACT

This study explores the potential of metaverse technologies for virtual education, comparing four platforms: Workrooms, Spatial, Mozilla Hubs, and Arthur. We evaluate factors like avatars, spatial arrangement, mobility, and functionalities' impact on concentration, usability, presence, and learning. While learning outcomes show no significant differences, substantial variations in the classroom experience emerge. Metaverses with limited movement and functionalities exhibit higher immersion, concentration, and presence. Moreover, avatars with enhanced facial expressiveness positively influence the overall learning experience.

Index Terms: Applied computing—Education—E-learning; Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Virtual reality

1 INTRODUCTION

Currently, there is a rising interest in virtual reality and the metaverse, technologies reshaping how we interact with the digital and physical world. The surge in virtual classes due to the Covid-19 pandemic has heightened interest in effective remote learning technologies, with tools like Mozilla Hubs (MHUB) being explored for educational purposes [4]. Given the potential of these technologies, virtual and augmented reality can significantly impact distance education, offering possibilities to enhance the learning experience, such as increased motivation, improved spatial representation of knowledge, and promotion of empirical learning [5]. Therefore, it is crucial to examine how the design of these environments influences learning and experience quality to optimize their use in education.

This research analyzes the experience and learning quality in four metaverse environments: Meta Workrooms (WKRM), Spatial (SPAT), MHUB, and Arthur (ARTH). Four different classes on "uses of extended reality" were conducted in these metaverses, targeting a university computer science related audience. Providing insights

into favorable and unfavorable characteristics for designing a learning environment in the metaverse, from avatar design to spatial organization and interaction techniques.

2 RELATED WORK

The use of Virtual Reality (VR) as an educational tool has been studied in academic literature. Some of these studied factors include the appearance of avatars, the position of students in the classroom, and peer behavior [3]. Additionally, optimal teacher proximity has been recognized as conducive to enhanced learning and attention [2, 5]. Moreover, there are studies that have investigated the influence of distractions or distracted behavior from peers on academic performance, emphasizing the importance of interruptions in shaping concentration and learning outcomes [1, 5].

Due to the Covid-19 pandemic, distance learning has explored various forms, including virtual reality classes and 3D virtual worlds like Minecraft and MHUB. Research has assessed the effectiveness of virtual reality features in enhancing interaction and student engagement, showing that 3D virtual classrooms provide a more immersive and effective learning experience compared to traditional classrooms [8]. Platforms like MHUB have proven useful for small groups, offering the ability to express oneself through body language, enhancing the teacher's perspective [4]. Additionally, the appearance of avatars in virtual interaction plays a crucial role, influencing engagement and performance [5, 7].

3 METAVERSE CHARACTERISTICS AND HYPOTHESIS

Regarding metaverse features, each platform offers unique characteristics. WKRM supports up to 16 participants with non-photorealistic upper body human avatars, light automatic facial expressions, and facial tracking with Meta Quest Pro. SPAT accommodates up to 50 participants, utilizing non-photorealistic full-body human avatars with mouth movement. It allows variable distance and free movement in a "Gallery Walk" environment (predefined by the instructors in this case) with predefined actions. MHUB supports up to 26 participants, employing non-photorealistic non-human avatars, allowing variable distance, free movement and emoticons in a "Gallery Walk" environment. ARTH has no participant limit, featuring photorealistic human avatars with variable distance and free movement, enabling interaction with objects in a free-exploration, single-room environment. Based on the above, the following hypotheses were selected:

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H1: Metaverses with avatars featuring more natural facial expressions will allow students to perceive less discomfort and more immersion.

H2: Applications that allow sitting or remaining in a fixed location are less distracting and favor concentration, improving learning outcomes.

H3: Applications with more interactions will be more distracting, negatively impacting attention and learning.

3.1 Methodology

Twenty-six students were allocated into four groups, each exposed once to every 10-minute class facilitated by two instructors, covering distinct metaverse topics: WKRM (Video Games), SPAT (Art and Design), MHUB (Education), and ARTH (Simulators).

Surveys assessed learning outcomes and overall experience, with a Pre-Learning Questionnaire (PRELQ) and Post-Learning Questionnaire (POSTLQ) evaluating understanding before and after sessions. The Post-Experience Questionnaire (PEQ) compiles feedback on perceived realism, presence, usability, and interest through 25 Likert-scale questions focusing on avatars, spatial organization, and interaction. Additionally, the Comparative Final Questionnaire (FCQ) tasks participants with ranking the four metaverse applications based on a statement, providing insights into the comparative quality of the experiences. Questionnaire administration follows a specific sequence: PRELQ before each session, POSTLQ and PEQ after each session, and FCQ once the individual has completed all four sessions.

The difference between PRELQ and POSTLQ results is compared using a Friedman. Additionally, a Tukey test was performed to assess whether there were significant differences between the deltas (The deltas are calculated by subtracting the average value of PRELQ from POSTLQ) among metaverses. The results of the FCQ undergo quantitative analysis through the application of an index. The FCQ index is the normalized average of the points obtained by the ranked position: if the metaverse was ranked in the first place, it receives 4 points and in the fourth place 1 point.

4 RESULTS AND DISCUSSION

The results of the Friedman Test reveal that there were no significant differences in learning scores before and after utilizing WKRM. Despite this, a slight, non-significant increase in post-test scores was observed for all metaverses, except MHUB. Similarly, the Tukey test did not yield significant differences; however, the data suggest that WKRM exhibited the highest learning delta, while MHUB displayed the lowest learning rate. The FCQ index results are as follows: WKRM: 0.96, SPAT: 0.59, ARTH: 0.48, and MHUB: 0.46. Notably, WKRM obtained the highest score, closely approaching the maximum value. Following WKRM is SPAT, while ARTH and MHUB occupy the last positions with values in close proximity.

WKRM excelled in avatar perception, particularly in aspects of naturalness, eye contact, and immersion according to the PEQ. This corroborates Hypothesis 1, as WKRM avatars demonstrated enhanced facial expressiveness. Additionally, instructors' utilization of the Meta Quest Pro allowed students to observe details such as facial expressions and gaze tracking, potentially fostering a more realistic connection. This may explain why, despite a similar appearance to WKRM, SPAT avatars, with limited expressiveness and eye movement, did not achieve comparable results. Nonetheless, SPAT supports Hypothesis 1 by securing the second position, presenting avatars with a caricature-like appearance but not highly expressive faces. ARTH, employing user photos for avatar facial textures, but consistently received low results, possibly due to the "uncanny valley" phenomenon [6]. This phenomenon aligns with Hypothesis 1, suggesting that avatars with less expressiveness and a non-caricature-like appearance generate less positive responses. In summary, the

research suggests that avatars with intermediate characteristics between realistic and caricature-like, coupled with heightened facial expressiveness, may be more effective for metaverse classes.

Observations from recordings indicated that in metaverses with more freedom of movement, individuals tended to explore more and pay less attention to the class. Additionally, PEQ results revealed that WKRM, with greater movement restrictions, provided a more natural sense of movement in the classroom, potentially explaining the positive concentration outcomes. Metaverses with more freedom of movement, such as MHUB, SPAT, and ARTH, were perceived as less natural and less conducive to concentration, supporting Hypothesis 2. The absence of legs in MHUB avatars may explain its better performance compared to ARTH and SPAT in this aspect. In summary, prioritizing the naturalness of movement is crucial for promoting immersion and concentration in the virtual classroom. Hypothesis 2 is partially fulfilled, suggesting that the naturalness of movement is more critical than restricting mobility.

In traditional classes, WKRM better adapted to virtual presence requirements, achieving superior interaction results. In contrast, ARTH presented distractions due to functionalities not considered in lesson design. This contrast partially supports Hypothesis 3, indicating that functionalities must align with class objectives. This principle applies not only to ARTH but also to SPAT and MHUB, which had more functions but proved distracting when not appropriately considered. In summary, Hypothesis 3 is partially correct, as effectiveness lies not in the quantity of functions but in their alignment with class objectives.

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